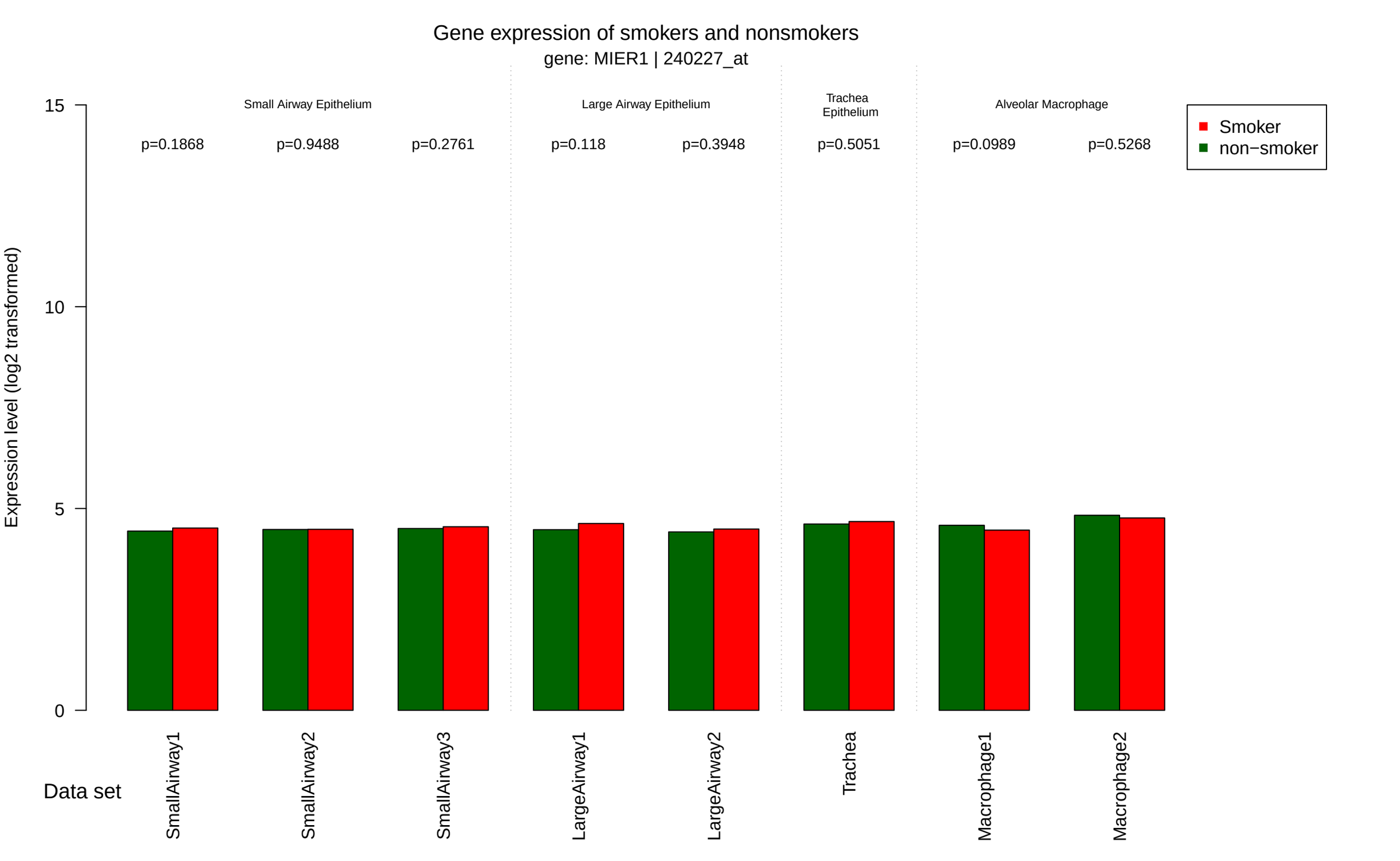
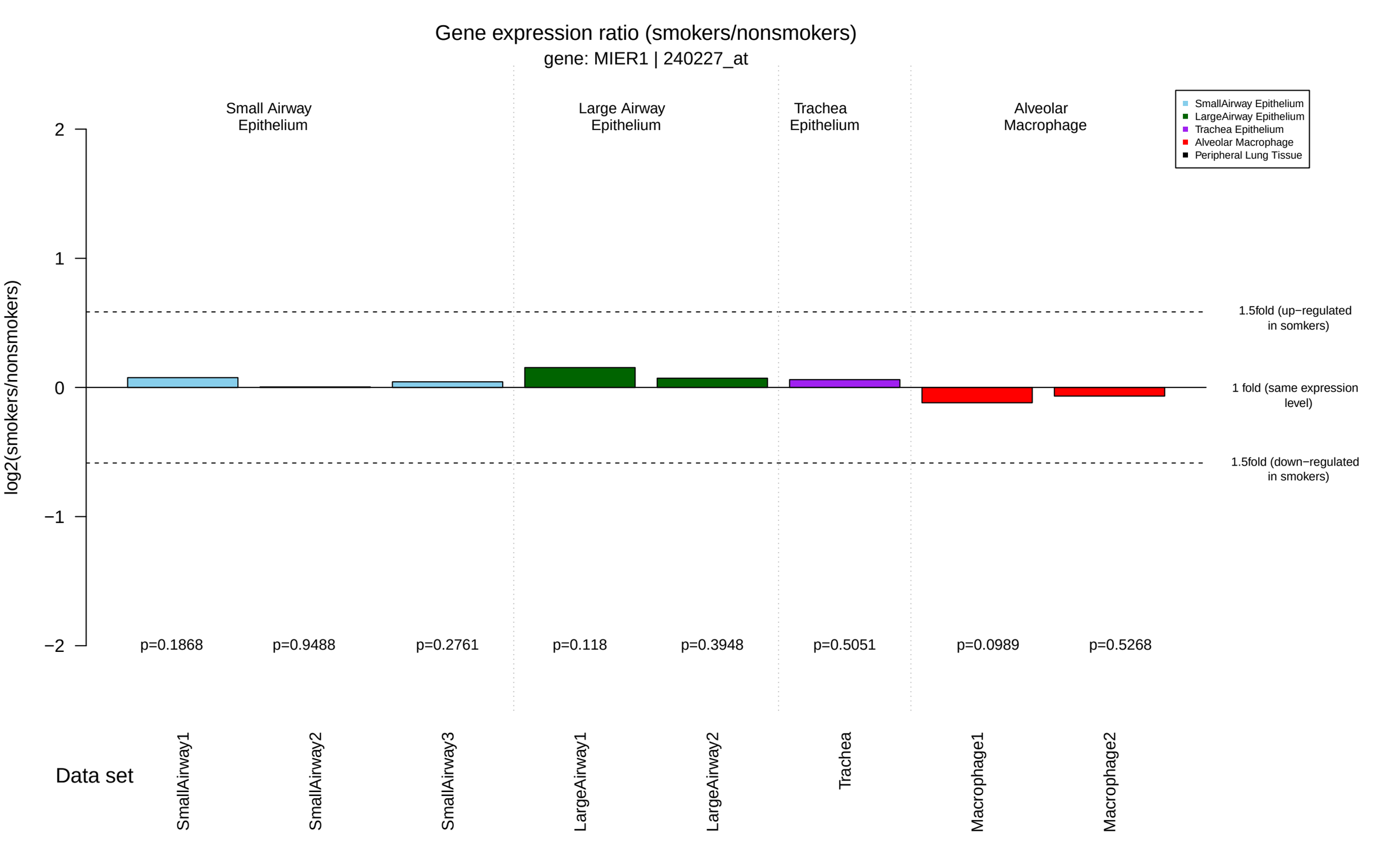
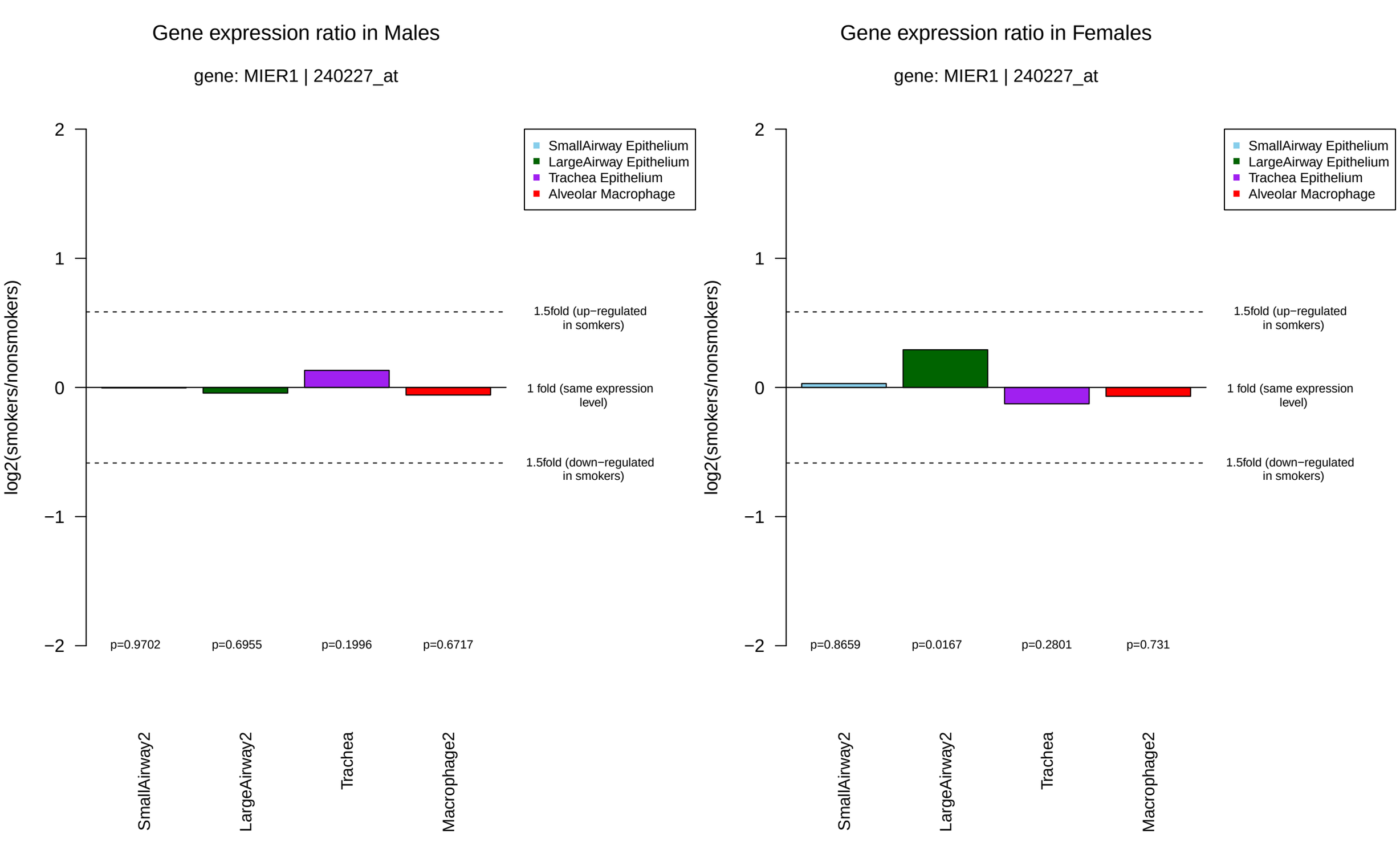
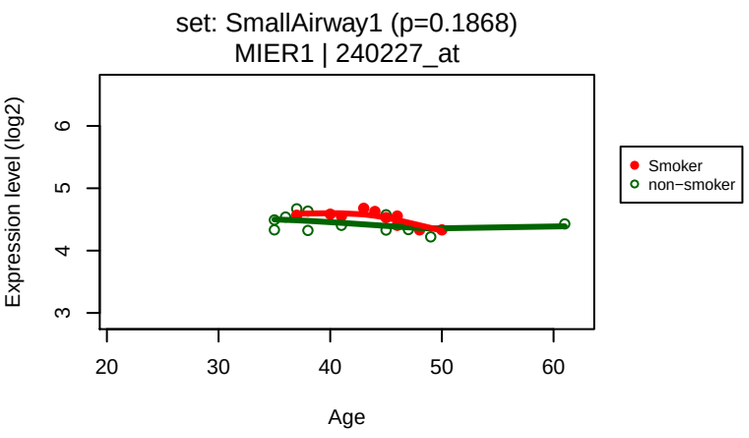
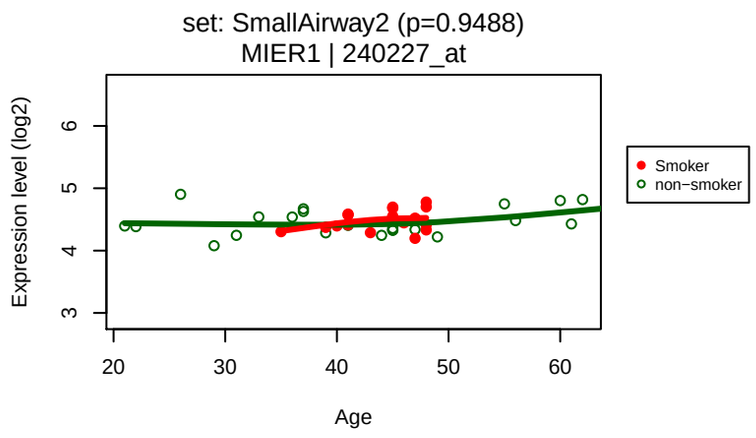
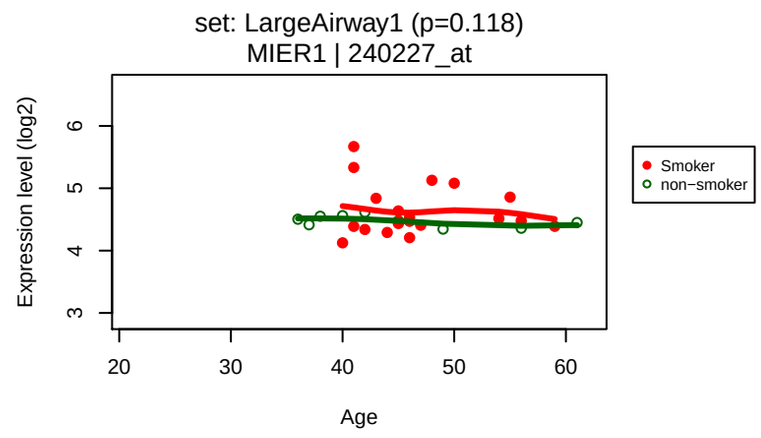

Gene expression of smokers and nonsmokers
gene: MIER1 | 240227_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1868
p=0.9488
p=0.2761
p=0.118
p=0.3948
p=0.5051
p=0.0989
p=0.5268
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

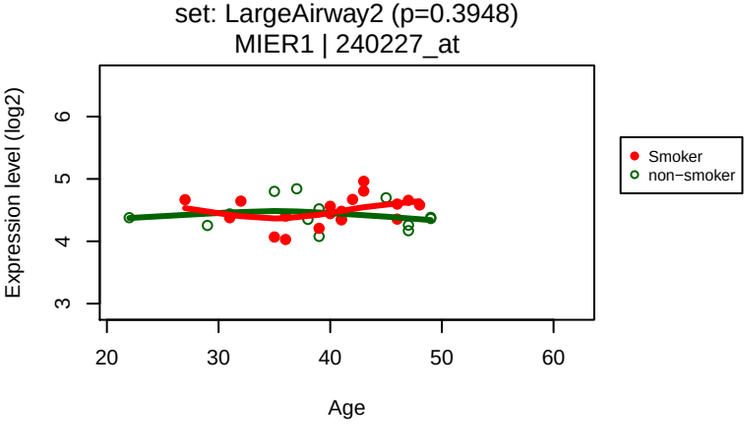

Gene expression ratio (smokers/nonsmokers)
gene: MIER1 | 240227_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1868
p=0.9488
p=0.2761
p=0.118
p=0.3948
p=0.5051
p=0.0989
p=0.5268
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

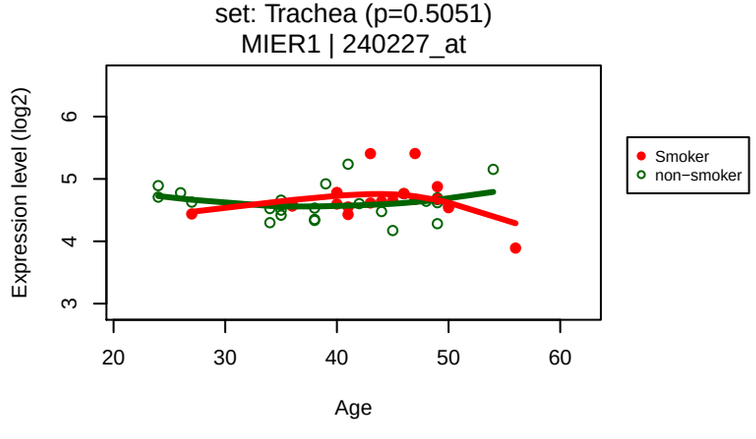

Gene expression ratio in Males
gene: MIER1 | 240227_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9702
p=0.6955
p=0.1996
p=0.6717
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: MIER1 | 240227_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8659
p=0.0167
p=0.2801
p=0.731
SmallAirway2
LargeAirway2
Trachea
Macrophage2

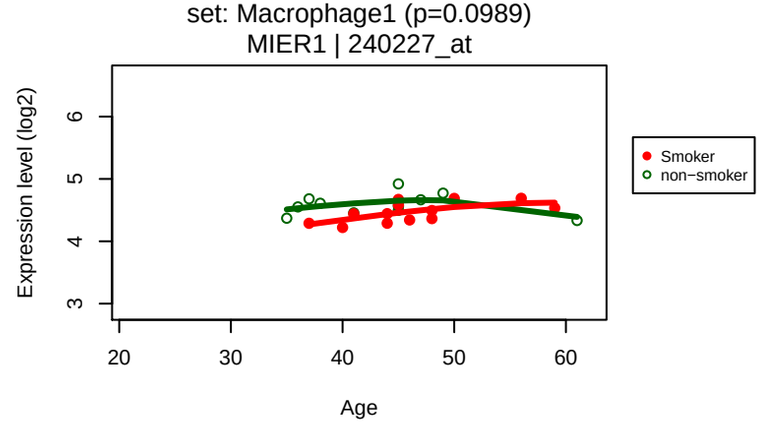

set: SmallAirway1 (p=0.1868)
MIER1 | 240227_at
Expression level (log2)
Age
Smoker
non−smoker

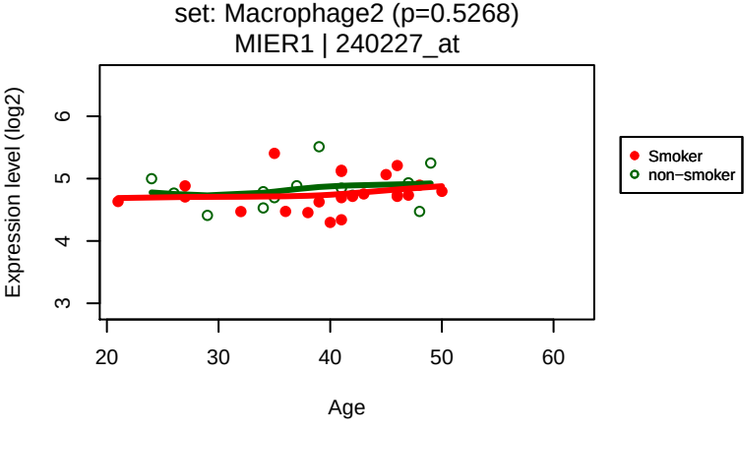

set: SmallAirway2 (p=0.9488)
MIER1 | 240227_at
Expression level (log2)
Age
Smoker
non−smoker

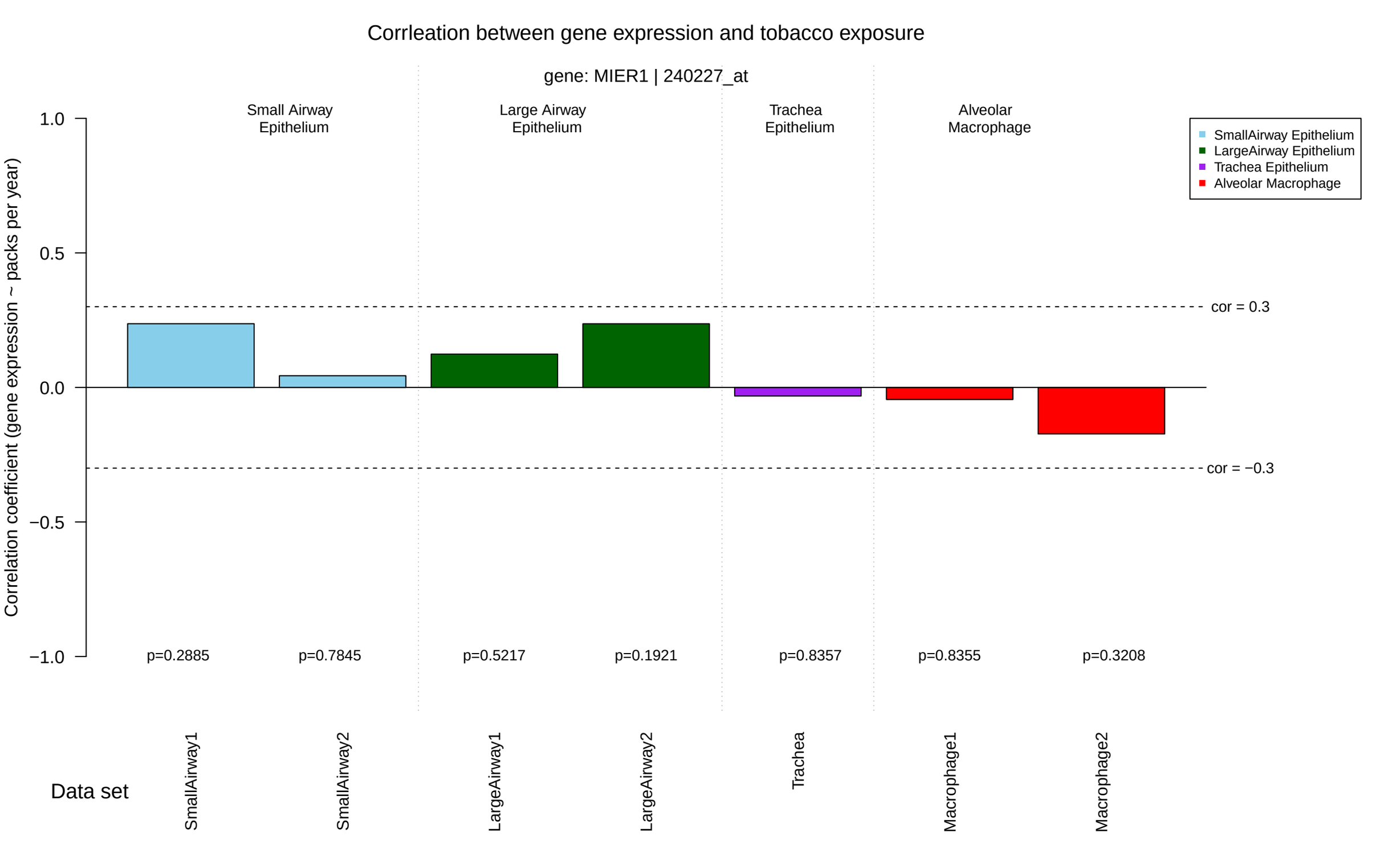

set: LargeAirway1 (p=0.118)
MIER1 | 240227_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3948)
MIER1 | 240227_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.5051)
MIER1 | 240227_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0989)
MIER1 | 240227_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.5268)
MIER1 | 240227_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: MIER1 | 240227_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2885
p=0.7845
p=0.5217
p=0.1921
p=0.8357
p=0.8355
p=0.3208
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2